

A Cloud Native Multi Modal AI Platform for Explainable Financial Fraud Detection

R26-IT-121

Project Proposal Report

Vidanaarachchi T.M (IT22192882)

B.Sc. (Hons) Degree in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

March 2026

A Cloud Native Multi Modal AI Platform for Explainable Financial Fraud Detection

R26-IT-121

Project Proposal Report

Supervised By: Mrs. Anjalie Gamage

B.Sc. (Hons) Degree in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

March 2026

Declaration

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education

institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgement is given in the text.

Name	Student ID	Signature
Vidanaarachchi T.M	IT22192882	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature

Abstract

Financial fraud within modern digital payment networks has evolved into highly sophisticated, multi-dimensional attacks. While advanced machine learning ensembles excel at detecting these anomalies, they inherently suffer from the “black box” problem. These models output abstract numerical probabilities without human readable reasoning, forcing compliance officers into time consuming manual investigations and resulting in a critically high Mean Time to Resolve (MTTR).

This research proposes a Fusion Engine and Generative Explainability component acting as the final orchestration and decision layer for the DeepSentinel multi modal fraud detection platform. The system introduces a Weighted Ensemble Meta Classifier that mathematically aggregates diverse risk vectors from upstream predictive models into a unified confidence score. Crucially, the research addresses the explainability gap by introducing a novel Typology Grounded Large Language Model (LLM) Forensic Reporting Layer. Instead of relying on free form text generation, this architecture utilizes Retrieval Augmented Generation (RAG) to query a local vector store of structured Financial Action Task Force (FATF) fraud typologies based on the fused numerical risk profile.

The LLM acts as a grounded forensic analyst, generating automated, audit traceable case reports where every investigative claim is anchored in specific mathematical evidence and documented crime patterns. Commercially, transforming abstract neural network scores into actionable, legally defensible forensic narratives significantly reduces investigation overhead. Socially, the system aligns with UN Sustainable Development Goal 16 by bridging the gap between advanced algorithmic detection and human centric regulatory compliance, enabling faster disruption of illicit financial flows.

Keywords: Retrieval Augmented Generation (RAG); Large Language Models (LLM); Ensemble Meta Classifier; Explainable AI (XAI); Digital Forensics.

Acknowledgement

The successful conceptualization and development of this research proposal would not have been possible without the guidance, support, and collaboration of several individuals and institutions.

First and foremost, I would like to express my sincere gratitude to my supervisor, Mrs. Anjalie Gamage, for her continuous support, technical insight, and invaluable feedback. Her expertise in the field has been instrumental in shaping the methodology and refining the architectural novelties of this research.

I extend my appreciation to the Department of Information Technology at the Sri Lanka Institute of Information Technology (SLIIT) and the Centre of Excellence for AI (COEAI) for providing the academic framework, computational resources, and rigorous research environment necessary to pursue a project of this scale.

Special thanks must be given to my DeepSentinel research team members. The realization of a true multi modal platform relies entirely on our seamless integration. The collaborative discussions regarding relational graph networks, behavioral profiling, and temporal rhythm analysis have significantly elevated the scope and real world applicability of this generative fusion component.

Table of Contents

1. INTRODUCTION	9
1.1 Background and Context	9
1.2 Literature Review	10
1.3 Research Gap	10
2. OBJECTIVE	11
2.1 Main Objective	11
2.2 Specific Objectives	11
3. METHODOLOGY	12
3.1 Research Approach	12
3.2 Technical Methods and Algorithms	12
3.3 Tools and Platforms	12
3.4 Data Collection and Ethical Considerations	13
3.5 Validation Strategy and Performance Metrics	13
4. HIGH LEVEL SYSTEM ARCHITECTURE	14
4.1 Overview of the Proposed System	14
4.2 System Components	14
4.3 Component Interaction	14
4.4 Data Flow Explanation	15
4.5 Scalability and Security Considerations	15
5. USER REQUIREMENTS	17
5.1 Stakeholder Identification	17
5.2 Functional Requirements	18
5.3 Non Functional Requirements	18
6. COMMERCIALIZATION PLAN	19
6.1 Target Market and Customer Persona	19
6.2 Value Proposition	19

6.3	Revenue Model.....	19
6.4	Go to Market Strategy	20
6.5	Pricing Strategy	20
6.6	Competitive Advantage	20
6.7	Intellectual Property Considerations	20
6.8	Cost Estimation	21
7.	BUDGET AND JUSTIFICATION.....	21
8.	WORK BREAKDOWN STRUCTURE (WBS)	22
9.	GANTT CHART	24
10.	REFERENCES LIST.....	24
11.	APPENDICES	26
11.1	Survey Instrument	26
11.2	Detailed diagrams	27
11.3	Preliminary UI mockups.....	28
11.4	Risk analysis	30
11.5	Ethical documents	31

List of Tables

Table 7.1 Estimated budget table	21
Table 8.1 work Breakdown Structure	22

List of Figures

Figure 4.2 High Level System Architecture	16
Figure 4.1 System Data Flow Diagram	17
Figure 9.1 Project Gantt Chart (March–November 2026)	23

List of Appendices

Appendix Figure A1. Detailed System Architecture Diagram.....	26
Appendix Figure A2. Detailed Data Flow Diagram	26
Appendix Figure A4. UI Mockup Batch Analytics Dashboard	27

List of Abbreviations

LIST OF ABBREVIATIONS

- AI: Artificial Intelligence
- AML: Anti Money Laundering
- API: Application Programming Interface
- AUROC: Area Under the Receiver Operating Characteristic Curve
- B2B: Business to Business
- CBSL: Central Bank of Sri Lanka
- CNN: Convolutional Neural Network
- CRO: Chief Risk Officer
- EDA: Exploratory Data Analysis
- FI GRL: Fast Inductive Graph Representation Learning
- GDP: Gross Domestic Product
- GDPR: General Data Protection Regulation
- GNN: Graph Neural Network
- GPU: Graphics Processing Unit
- JSON: JavaScript Object Notation
- LLM: Large Language Model
- MLP: Multi Layer Perceptron

MTTR: Mean Time to Resolve

OOM: Out Of Memory

PII: Personally Identifiable Information

PoC: Proof of Concept

PyG: PyTorch Geometric

SaaS: Software as a Service

SDG: Sustainable Development Goals

TCN: Temporal Convolutional Network

UNODC: United Nations Office on Drugs and Crime

VAE: Variational Autoencoder

VRAM: Video Random Access Memory

WBS: Work Breakdown Structure

1. INTRODUCTION

1.1 Background and Context

The global financial landscape has completed a rapid transition toward real time digital payment ecosystems. While this shift enhances economic efficiency, it has simultaneously accelerated the sophistication of organized financial crime. Criminal syndicates now deploy highly complex, multi dimensional attacks, utilizing algorithmic mule networks and rapid temporal structuring to bypass standard security thresholds. In response, modern anti money laundering (AML) architectures have shifted away from simple rule based heuristics toward advanced, multi modal deep learning systems. These contemporary platforms deploy specialized models such as Graph Neural Networks for relational topology and Variational Autoencoders for behavioral deviation to analyze massive volumes of transactional data simultaneously.

However, this algorithmic evolution has created a critical secondary crisis in the financial security pipeline: the crisis of explainability. When modern deep learning models process transactions, they typically function as opaque “black boxes”. They output abstract risk scores indicating a high probability of fraud, but they offer no transparent, human readable reasoning detailing why that specific mathematical decision was made.

In heavily regulated commercial banking environments, a raw probability score is legally and operationally insufficient. A compliance officer cannot authorize an account freeze, confiscate funds, or submit a Suspicious Activity Report (SAR) to regulatory bodies based solely on a neural network's numerical output. Human investigators must understand the underlying narrative of the crime. Currently, investigators are forced to bridge the gap between abstract AI alerts and actionable evidence manually, reviewing database logs to reverse engineer what the AI detected. This manual translation process creates severe operational bottlenecks, driving up the Mean Time to Resolve (MTTR) and allowing illicit funds to escape before security teams can legally justify an intervention.

This research proposes a Fusion Engine and Generative Explainability module as the final decision intelligence layer for the DeepSentinel multi modal security platform. Operating as the central orchestrator, this component receives asynchronous intelligence from upstream specialized analytical modules. It fuses these isolated signals into a unified confidence score and leverages a Retrieval Augmented Generation (RAG) framework to automatically translate complex multi modal anomalies into comprehensive, mathematically grounded forensic investigation reports.

Statistical Context and SDG Alignment: The operational cost of the “black box” problem is severe; industry metrics indicate that manual alert investigation costs financial institutions billions annually, with false positives draining compliance resources. This research directly aligns with **United Nations Sustainable Development Goal 16 (Peace, Justice and Strong Institutions)** by providing transparent, actionable intelligence to regulatory bodies combating organized financial crime. Furthermore, it supports **SDG 9 (Industry, Innovation, and Infrastructure)** by integrating state of the art Generative AI safely into critical banking infrastructure.

1.2 Literature Review

The challenge of combining multiple fraud detection models and interpreting their outputs is well documented in recent literature. Stacking ensemble meta classifiers are established mechanism for score fusion. For example, the Integrated Multistage Ensemble Machine Learning (IMEML) framework published in the Journal of Big Data (November 2024)[1] and the Risk Adaptive Bayesian Ensemble Method (RABEM) applied specifically to the PaySim dataset (October 2025) [2] successfully demonstrate how combining separate predictive signals via meta learners increases overall detection accuracy. However, while ensemble methods improve predictive accuracy, they actively compound the black box problem by making the final decision even harder to trace across multiple overlapping algorithms.

Traditional attempts to solve this transparency issue rely heavily on Explainable AI (XAI) techniques like SHAP (SHapley Additive exPlanations) or LIME. A 2025 IEEE Access study on stacking ensemble fraud detection utilized SHAP and LIME to assign feature importance scores [3]. Yet, as confirmed by a 2024 IEEE Access systematic review of XAI in finance [4], SHAP produces numerical attribution metrics that do not offer transparent or user-friendly reasoning for legal or regulatory reporting. A compliance officer cannot submit a SHAP feature plot as a legal case report.

Consequently, the industry is rapidly pivoting toward Large Language Models (LLMs) to generate natural language reasoning. Prompt engineering techniques for financial decision making have been extensively reviewed [10], and commercial systems like KPMG's VertexAI pipeline currently utilize LLMs to summarize standard financial filings and generate text reports. Similarly, Taktile implemented LLMs as investigative reasoning engines in production fintech systems in 2025. However, these existing LLM forensic tools [5] operate almost entirely on raw textual data (e.g. call transcripts, corporate reports) and remain entirely disconnected from the numerical risk vectors produced by multi modal deep learning detection pipelines. Furthermore, applying pure prompt engineering on LLMs without factual grounding is highly susceptible to generative hallucination. An April 2025 digital forensics paper documented real legal cases where LLM generated forensic evidence was deemed inadmissible in court because its reasoning was untraceable to source data, explicitly calling for RAG grounded systems as critical future work [7].

1.3 Research Gap

Despite rapid advancements in both deep learning ensembles and Generative AI, a comprehensive 2025 survey of financial AI papers confirms a critical architectural gap: no existing system combines a multi modal deep learning fraud ensemble (Graph + Behavioral + Temporal) with a Retrieval Augmented Generation (RAG) layer [9] that grounds an LLM's forensic narrative in a structured fraud typology knowledge base.

First, there is a clear gap regarding the **Forensic Translation of Deep Learning Vectors**. Current multi modal architectures either produce no explanation or rely strictly on numerical attribution (SHAP) that is not user friendly or legally actionable for human investigators. They fail to translate complex matrix outputs and multi dimensional risk signals into coherent, actionable forensic narratives.

Second, a critical gap exists regarding **Grounded LLM Output in Financial Analytics**. Existing LLM applications in fraud detection predominantly process text documents and act as free-form text generators, creating severe legal admissibility risks due to hallucination. A January 2026 ScienceDirect survey identified "forensic-ready LLM architectures with traceable outputs" as a critical open research gap [6]. Furthermore, recent surveys confirm that standard RAG architectures struggle to integrate structured domain-specific knowledge bases, reinforcing the need for a typology-grounded retrieval layer [11].

This research directly addresses these gaps by introducing a Typology Grounded LLM Forensic Analyst. Instead of free form text generation, the proposed architecture first maps incoming multi modal risk scores to a local vector store of documented FATF (Financial Action Task Force) fraud typologies. The LLM then utilizes this retrieved, structured context as a factual anchor to generate a traceable, evidence linked case report, effectively bridging the gap between high dimensional AI detection and human readable digital forensics.

2. OBJECTIVE

2.1 Main Objective

To design, implement, and evaluate a Weighted Ensemble Fusion Engine that aggregates multi modal risk scores into a unified Fraud Confidence Score, and to architect a Typology Grounded LLM Forensic Reporting Layer utilizing Retrieval Augmented Generation to produce audit traceable, natural language investigation reports.

2.2 Specific Objectives

- To develop an asynchronous orchestration script using FastAPI capable of receiving continuous transaction data and dispatching parallel internal API requests to upstream graph, behavioral, and temporal analytical modules without blocking system execution.
- To implement a Meta Classifier stacking layer utilizing Logistic Regression to fuse distinct probability outputs into a final, optimized confidence score based on calibrated validation weights.
- To construct a local, privacy preserving vector knowledge base (ChromaDB) containing structured definitions of known financial crime typologies drawn from the Financial Action Task Force (FATF) and academic taxonomies.
- To engineer a Chain of Evidence prompt architecture that forces a Large Language Model to construct forensic narratives strictly utilizing the numerical sub model findings and the retrieved FATF context, thereby eliminating generative hallucination.
- To design and deploy a real time interactive dashboard (Streamlit) that visualizes per modality risk gauges alongside the generated text reports, providing compliance investigators with an immediate, actionable interface

3. METHODOLOGY

3.1 Research Approach

This study follows a quantitative design and development research approach, specifically focused on systems engineering and natural language processing (NLP). The core problem is that deep learning ensembles output abstract numerical arrays. The engineering approach involves building a deterministic integrated pipeline that mathematically balances those numbers, searches a vector database for a matching crime profile, and restricts a language model to translate the findings into a legal forensic format. The system is developed as an independent orchestration layer that actively pulls data from the specialized upstream models, demonstrating its viability as a production grade decision support tool.

3.2 Technical Methods and Algorithms

The system relies on three primary technical methods functioning in strict sequence:

- **Weighted Ensemble Meta Classification:** A Logistic Regression stacking layer acts as the meta classifier. It ingests the distinct probability scores from the graph, behavioral, and temporal models and applies learned weights to output a final Fraud Confidence Score. This ensures the system does not simply average the inputs, but statistically trusts the most accurate models for specific fraud types based on validation performance.
- **Retrieval Augmented Generation (RAG) [9]:** Instead of allowing an LLM to generate free form text based on numerical inputs, the system converts the incoming risk profile into a semantic search query. It uses cosine similarity search to find the closest matching financial crime pattern (e.g. Smurfing, Layering, Account Takeover) residing in a pre built vector database.
- **Chain of Evidence Prompting:** The retrieved crime definition and the raw numerical risk scores are combined into a strict instructional prompt template. The language model is explicitly restricted to citing only the provided numbers and the retrieved typology text, anchoring the final narrative to verified facts.

3.3 Tools and Platforms

The development environment relies on Python 3.11 and several highly specialized frameworks chosen for their efficiency and local execution capabilities:

- **Pipeline Orchestration:** Native asynchronous Python libraries (httpx and asyncio) will manage the internal RESTful API routing, allowing the system to dispatch parallel requests to the sub models and handle computational timeouts without halting the entire pipeline.
- **Data Fusion:** scikit learn handles the Logistic Regression ensemble layer due to its lightweight and highly interpretable nature.

- **Knowledge Base & Embeddings:** ChromaDB serves as the local vector store. It is populated using the sentence transformers library (specifically the all MiniLM L6 v2 model), which provides high accuracy semantic search while running entirely offline.
- **Language Model:** The generative reporting will be processed via the Gemini API (for rapid demonstration) or a locally hosted Llama 3 instance via Ollama (for offline, privacy compliant execution).
- **User Interface:** Streamlit combined with Plotly will power the real time forensic dashboard, visualizing the fused risk gauges and textual reports.

3.4 Data Collection and Ethical Considerations

The meta classifier will be trained and calibrated using the validation splits from the PaySim dataset [8], a publicly available financial simulator. Because PaySim is entirely synthetic, it contains no Personally Identifiable Information (PII), eliminating major privacy and compliance barriers. The fraud typology knowledge base is constructed strictly using publicly available guidance documents and definitions from regulatory bodies like FATF and established academic taxonomies. By utilizing local vector storage (ChromaDB) and supporting offline language models (Ollama), the architecture ensures that sensitive, simulated transactional risk vectors do not need to be transmitted to external, third party AI servers, preserving institutional data security.

3.5 Validation Strategy and Performance Metrics

Standard machine learning accuracy metrics do not apply to generative text. Therefore, validation requires a multi tiered approach covering each sequential step of the pipeline:

- **Fusion Accuracy:** The meta classifier will be evaluated using Precision, Recall, and F1 score against the PaySim validation set to mathematically verify that the fused score outperforms the isolated upstream models.
- **Retrieval Testing:** The RAG database will undergo targeted querying to verify that specific numerical inputs correctly retrieve the expected conceptual definition (e.g. “Smurfing”).
- **Ablation Study (Legal Admissibility):** The core novelty will be validated by comparing reports generated by pure prompt engineering against reports generated by the RAG grounded architecture. A Legal Admissibility Rubric will measure the reduction in generative hallucination, failing any output that invents transaction details or fails to cite the retrieved FATF context.

4. HIGH LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed System

The Fusion Engine and Generative Explainability component functions as the terminal orchestration and decision layer of the overall DeepSentinel platform. The architecture is designed to transition raw predictive data into structured human intelligence. It operates asynchronously, pulling numerical risk probabilities via internal RESTful APIs from independent upstream anomaly detectors (graph, behavioral, and temporal models). The system mathematically fuses these inputs to determine an overall confidence score, retrieves matching regulatory definitions from a local database, and constructs a legally defensible forensic report using a constrained Large Language Model.

4.2 System Components

The system consists of five highly specialized, interconnected analytical modules:

- **Asynchronous Integration Controller:** A Python based FastAPI routing script that serves as the central traffic controller. It dispatches parallel requests to upstream analytical models and contains fallback logic to handle potential model timeouts without breaking the pipeline.
- **Weighted Ensemble Meta Classifier:** A Logistic Regression stacking layer that ingests the three distinct predictive scores and applies pre calibrated weights to compute a single, optimized Fraud Confidence Score.
- **FATF Fraud Typology Knowledge Base:** A local ChromaDB vector store populated with embedded, structured definitions of known financial crime patterns (e.g. Layering, Mule Networks, Smurfing) drawn from the Financial Action Task Force.
- **Typology Grounded LLM Forensic Analyst:** The core generative reasoning engine. It utilizes a Retrieval Augmented Generation (RAG) framework and a Chain of Evidence prompt template to restrict the LLM to writing reports based exclusively on the numerical inputs and the retrieved FATF database context.
- **Interactive Forensic Dashboard:** A Streamlit based user interface that visualizes the risk metrics via gauge charts and displays the final generated text report for the human investigator to review.

4.3 Component Interaction

The architecture follows a strict, sequential pipeline. The Integration Controller triggers the operation by fetching scores from the upstream models. Once the scores arrive, they are passed directly to the Meta Classifier to calculate the final confidence probability. This multi dimensional risk profile is then converted into a search query and sent to the ChromaDB Knowledge Base. The database returns the closest matching crime typology. Finally, the raw numerical scores and the retrieved text definition are packaged together and sent to the LLM, which formats the data into the final forensic narrative displayed on the Streamlit dashboard.

4.4 Data Flow Explanation

Incoming Transaction Data → Parallel Asynchronous API Calls to Upstream Models → Receive 3x Probability Scores → Logistic Regression Meta Classifier → Compute Fused Fraud Confidence Score → Cosine Similarity Search in ChromaDB → Retrieve FATF Typology Context → Inject Scores and Context into Chain of Evidence Prompt → LLM Processing → Output Grounded Forensic Narrative → Streamlit Dashboard Display.

4.5 Scalability and Security Considerations

Scalability: The integration controller is built using asynchronous Python libraries (asyncio). This ensures that if one upstream detection model experiences heavy load and delays its calculation, the Fusion Engine's main thread is not blocked. Furthermore, the controller is engineered with fallback logic; if a sub model times out completely, the system proceeds with the available scores and flags the missing modality in the final report.

Security & Privacy: Financial transaction data is highly sensitive. To mitigate the risks of exposing institutional data to public AI networks, the FATF Knowledge Base is hosted entirely locally using ChromaDB and open source embedding models (sentence transformers). For enterprise deployment, the architecture supports entirely offline, locally hosted language models (e.g. Llama 3 via Ollama) to ensure that no transaction metadata ever leaves the internal pipeline environment.

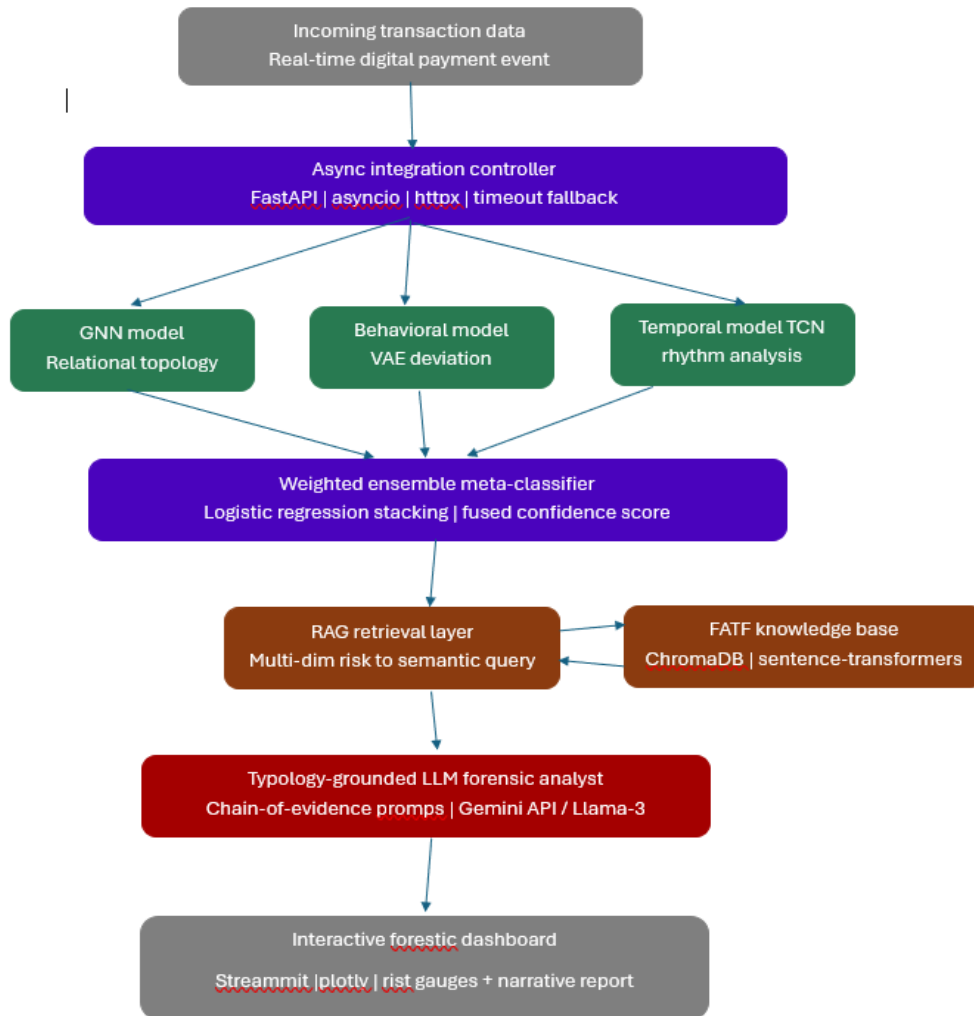


Figure 4.2 High Level System Architecture

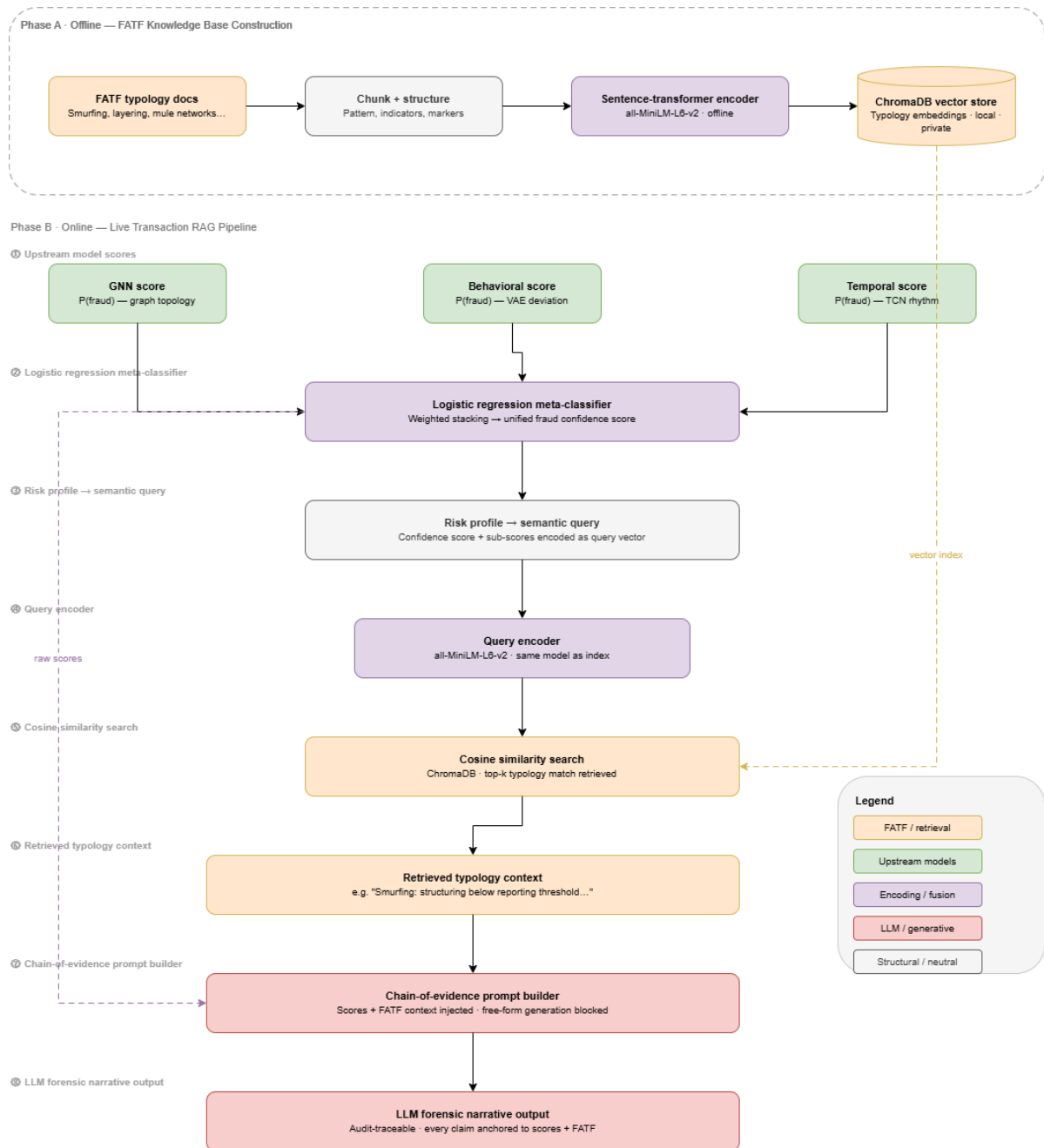


Figure 4.1 System Data Flow Diagram

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The Fusion Engine and Generative Explainability module serves as the primary user facing intelligence layer of the DeepSentinel platform. Therefore, its stakeholders include both the operational end users who read the reports and the technical teams who maintain the pipeline:

- **Anti Money Laundering (AML) Investigators:** The primary operational users. They interact directly with the Streamlit dashboard. They rely entirely on the LLM generated forensic narratives to understand the context of an alert, allowing them to rapidly decide whether to escalate a case or dismiss it as a false positive.
- **Compliance and Legal Officers:** Responsible for regulatory reporting. They require the system's output to be deterministic, legally admissible, and free of generative hallucination. They rely on the RAG architecture to ensure that every AI claim is backed by a verified Financial Action Task Force (FATF) typology.
- **Data Science / Risk Engineering Teams:** The technical stakeholders who deploy and monitor the system. They require the Logistic Regression meta classifier to be easily retrainable so they can adjust the ensemble weights if one of the upstream models starts underperforming.

5.2 Functional Requirements

To successfully orchestrate the multi modal pipeline and generate grounded reports, the system must meet the following functional requirements:

- **FR1 (Asynchronous Orchestration):** The system shall establish an integration controller capable of dispatching parallel internal API calls to the upstream graph, behavioral, and temporal models to collect their respective probability scores.
- **FR2 (Graceful Degradation & Fallback):** If one or more upstream models time out or fail to respond, the system shall not crash; it must proceed with the available scores, adjust the meta classifier input, and explicitly flag the missing data modality within the final generated report.
- **FR3 (Ensemble Fusion):** The system shall utilize a calibrated Logistic Regression stacking layer to mathematically combine the incoming risk vectors into a single, unified Fraud Confidence Score.
- **FR4 (Context Retrieval):** The system shall convert the fused multi dimensional risk profile into a semantic query and execute a cosine similarity search against a local ChromaDB vector store to retrieve the closest matching FATF fraud typology.
- **FR5 (Grounded Generation):** The system shall utilize a Chain of Evidence prompt template to force the integrated Large Language Model (LLM) to generate a structured forensic case report citing strictly the numerical scores and the retrieved FATF database context.
- **FR6 (Interactive Visualization):** The system shall render a real time Streamlit dashboard that displays the per modality risk gauges, the overall confidence score, the retrieved FATF match percentage, and the final LLM generated narrative.

5.3 Non Functional Requirements

To ensure the system is viable for a secure financial environment, it must adhere to the following non functional constraints:

- **Traceability and Admissibility (Zero Hallucination):** The generative reporting layer must strictly adhere to the provided context. The system must not invent transaction amounts, hallucinate account numbers, or propose fraud typologies that do not exist within the local FATF database.
- **Data Privacy and Security:** The system must protect simulated financial metadata by ensuring the knowledge base (ChromaDB) operates entirely locally. For offline enterprise testing, the architecture must support local execution of the LLM via frameworks like Ollama to prevent data leakage to public APIs.
- **Performance and Latency:** The asynchronous orchestration script must collect the upstream scores in under 500 milliseconds. The entire RAG retrieval and LLM generation process must complete within a reasonable operational window (e.g. under 5 seconds) to ensure the Streamlit dashboard updates in near real time for the investigator.
- **Interoperability:** The backend integration controller must strictly adhere to predefined JSON schemas to ensure flawless data parsing when receiving payloads from the independent analytical modules developed by the other research members.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed Fusion Engine and Generative Explainability module—and the broader DeepSentinel platform—comprises financial institutions operating within high volume digital payment ecosystems. This includes Neo banks, digital wallet providers, and traditional commercial banks transitioning to real time payment architectures. The primary customer persona is the Chief Risk Officer (CRO) or the Head of Anti Money Laundering (AML) Compliance. They require a system that eliminates the massive overhead of manual investigations.

6.2 Value Proposition

The core value proposition of this generative component is the complete elimination of the “Black Box” problem in financial AI. Currently, highly accurate deep learning models are bottlenecked by human investigators who must manually reverse engineer why an alert was generated. By translating complex, multi modal risk vectors into structured, human readable forensic narratives grounded in verified Financial Action Task Force (FATF) typologies, this system significantly reduces the Mean Time to Resolve (MTTR) for critical fraud alerts.

6.3 Revenue Model

The system is designed for commercialization as a B2B Enterprise software solution. Due to strict data privacy regulations in the financial sector, the platform will be offered primarily

as an Enterprise On Premise deployment. The integrated machine learning pipeline, including the local ChromaDB vector store and offline LLM instances, will be deployed directly within the bank's secure internal IT infrastructure. Revenue generation will rely on an annual enterprise licensing fee, supplemented by a volume based integration tier tied to the number of transactions processed by the orchestration controller daily.

6.4 Go to Market Strategy

Market entry will be executed through strategic Proof of Concept (PoC) pilot programs with mid tier fintech platforms. The primary adoption strategy is a "Shadow Mode" deployment, where the DeepSentinel reporting layer runs concurrently alongside the institution's legacy rule based dashboard without actively blocking transactions. After a 30 day evaluation window, the commercial pitch will focus on the "Explainability Delta," presenting the client with standard alerts generated by their legacy system alongside the comprehensive, multi page forensic reports generated by our Typology Grounded LLM.

6.5 Pricing Strategy

A tiered subscription pricing strategy ensures the system remains accessible to scaling fintech startups while capturing maximum value from major commercial banks:

- **Standard Tier:** Geared toward mid sized digital wallets. Includes the basic ensemble meta classifier, standard RAG retrieval, and cloud based LLM generation via secured enterprise APIs (e.g. Gemini API).
- **Enterprise Air Gapped Tier:** Designed for large commercial banks with strict data localization laws. Includes fully local execution of the Large Language Model (e.g. Llama 3 via Ollama) and a dedicated, locally hosted FATF ChromaDB instance, ensuring zero transactional metadata ever leaves the bank's internal network.

6.6 Competitive Advantage

This component establishes a highly defensible competitive moat against both legacy vendors and modern AI startups. Legacy systems rely on static rules and offer no generative explainability. Modern competitors primarily apply generative AI to summarize raw text documents (corporate filings, call transcripts). Our competitive advantage is the Typology Grounded RAG Architecture connected directly to Deep Learning Risk Vectors. Unlike competitors whose LLMs are susceptible to hallucination, our system forces the LLM to anchor every claim to a retrieved FATF crime pattern and a specific mathematical output, guaranteeing audit traceable reporting.

6.7 Intellectual Property Considerations

The core intellectual property (IP) resides in the proprietary orchestration logic and the prompt engineering architecture. Specifically, the programmatic logic converting multi

modal probability matrices into semantic search queries, and the exact structure of the Chain of Evidence prompt template, constitute proprietary trade secrets. The specific structural application of combining deep learning ensembles with RAG for financial digital forensics holds significant patentable potential.

6.8 Cost Estimation

The estimated cost for developing, integrating, and evaluating the Fusion Engine and Generative Explainability prototype over the academic year is Rs. 43,000. This budget is allocated strictly toward the computational resources required for local vector database hosting, ensemble model validation, and high tier LLM API access.

7. BUDGET AND JUSTIFICATION

Description	Total Amount (Rs.)	Justification
Cloud Compute & LLM API Access	15,000	Essential for accessing high tier enterprise language models (e.g. Gemini API) during the iterative testing of the Chain of Evidence prompt architecture, and for handling the computational overhead of local Llama 3 execution during offline testing.
Integration & Dashboard Server	8,000	Hosting the FastAPI asynchronous orchestration script and the real time Streamlit dashboard to simulate continuous transaction flows and ensure sub 500ms API routing between the upstream analytical modules.
Vector Database & Storage Services	6,000	Required for hosting the local ChromaDB vector store, processing the sentence transformer embeddings, and securely storing the generated forensic report audit logs.
High Bandwidth Internet and Utilities	4,000	Allocated costs for high bandwidth connectivity required for continuous API

		synchronizations, downloading massive localized language models
Human effort	5,000	Estimated allowance for student researcher time allocated to dataset preparation, model training, expert consultation, and evaluation activities
Contingency	4,000	Buffer for unexpected cloud compute spikes during the final end to end ablation studies or additional API token costs during the Legal Admissibility testing phase.
Total Budget	43,000	

Table 7.1 Estimated budget table

8. WORK BREAKDOWN STRUCTURE (WBS)

Task ID	Task Description	Timeline	Deliverable
T1	Conduct literature review on ensemble meta classifiers, Retrieval Augmented Generation (RAG), and LLM forensic translation.	March	Literature review report
T2	Configure Python development environment, FastAPI orchestration skeleton	March April	Environment setup
T3	Implement Logistic Regression Meta Classifier and optimize fusion weights using the PaySim validation dataset.	April	Calibrated meta classifier module
T4	Construct FATF Fraud Typology Knowledge Base and initialize the local ChromaDB vector store.	May	Configured vector database

T5	Embed typology definitions and validate cosine similarity retrieval accuracy.	May – June	Semantic retrieval script
T6	Develop Typology Grounded LLM integration and engineer the Chain of Evidence prompt architecture.	June	LLM prompt template
T7	Test RAG grounded LLM report generation using mock multi modal risk vectors to verify factual anchoring.	July	Preliminary forensic reports
T8	Integrate real analytical module APIs (graph, behavioral, temporal) and replace mock generators for end to end pipeline.	August	Fully integrated Fusion Engine
T9	Develop interactive Streamlit forensic dashboard with per modality risk gauges and report visualization.	September	Streamlit dashboard prototype
T10	Conduct ablation studies using the Legal Admissibility Rubric and perform pipeline latency stress testing.	October	Evaluation and testing report
T11	Prepare final thesis chapters, system documentation, and defense presentation.	November	Final research report & slides

Table 8.1 work Breakdown Structure

9. GANTT CHART

The Gantt chart below presents the planned project timeline from March to November 2026, aligned with the eleven tasks defined in the Work Breakdown Structure.

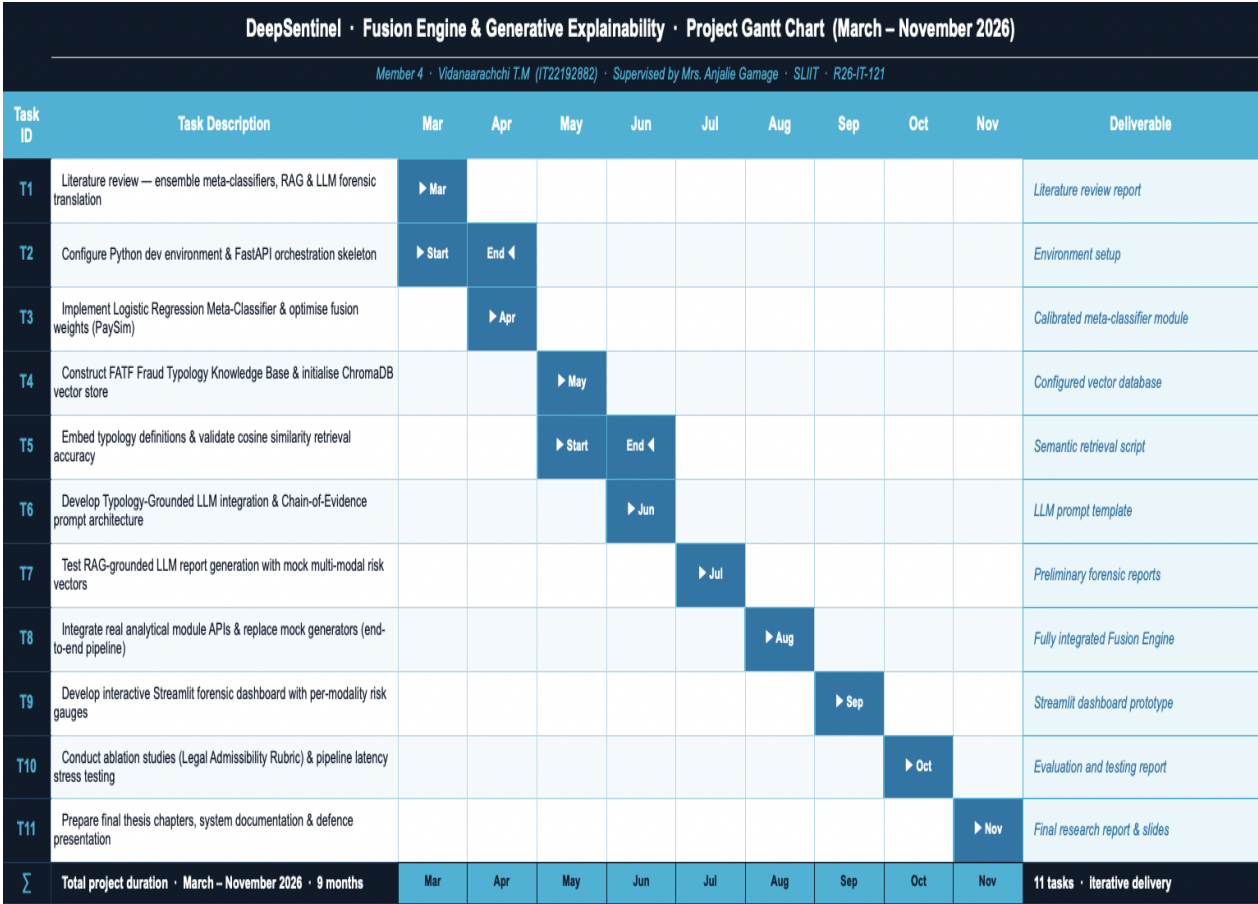


Figure 9.1 Project Gantt Chart (March–November 2026)

10. REFERENCES LIST

[1] M. A. Talukder, M. M. Islam, M. A. Uddin, K. F. Hasan, S. Sharmin, S. A. Alyami, and M. A. Moni, “An Integrated Multistage Ensemble Machine Learning Model for Fraudulent Transaction Detection,” Journal of Big Data, vol. 11, no. 168, Nov. 2024, doi: 10.1186/s40537-024-00996-5

- [2] F. A. Almarshad, M. Zakariah, G. A. Gashgari, and T. Vaiyapuri, "RABEM: Risk Adaptive Bayesian Ensemble Model for Fraud Detection," *Scientific Reports*, vol. 15, no. 1, p. 36796, Oct. 2025, doi: 10.1038/s41598-025-20651-0.
- [3] N. Damanik and C.-M. Liu, "Advanced Fraud Detection: Leveraging K-SMOTEENN and Stacking Ensemble to Tackle Data Imbalance and Extract Insights," *IEEE Access*, vol. 13, pp. 10356–10370, 2025, doi: 10.1109/ACCESS.2025.3528079."
- [4] T. Martins, A. M. De Almeida, E. Cardoso, and L. Nunes, "Explainable Artificial Intelligence (XAI): A Systematic Literature Review on Taxonomies and Applications in Finance," *IEEE Access*, vol. 12, pp. 618–629, 2024, doi: 10.1109/ACCESS.2023.3347028.
- [5] A. Wickramasekara, F. Breitingner, and M. Scanlon, "Exploring the Potential of Large Language Models for Improving Digital Forensic Investigation Efficiency," *Computers & Security*, vol. 150, p. 104160, Jan. 2025, doi: 10.1016/j.cose.2025.104160.
- [6] M. Chernyshev, Z. Baig, N. Syed, R. Doss, and M. Shore, "Large Language Models in Digital Forensics: Capabilities, Challenges and Future Directions," *Forensic Science International: Digital Investigation*, Jan. 2026, doi: 10.1016/j.fsidi.2025.301830.
- [7] Z. Yin, Z. Wang, W. Xu, J. Zhuang, P. Mozumder, A. Smith, and W. Zhang, "Digital Forensics in the Age of Large Language Models," *arXiv preprint*, arXiv:2504.02963v1, Apr. 2025. [Online]. Available: <https://arxiv.org/abs/2504.02963>
- [8] E. A. Lopez Rojas, A. Elmir, and S. Axelsson, "PaySim: A Financial Mobile Money Simulator for Fraud Detection," in *Proc. 28th European Modeling and Simulation Symposium (EMSS)*, Larnaca, Cyprus, pp. 249–255, 2016.
- [9] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval Augmented Generation for Knowledge Intensive NLP Tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.

[10] Y. Li, S. Wang, H. Ding, and H. Chen, “Large Language Models in Finance: A Survey,” in Proc. 4th ACM International Conference on AI in Finance (ICAIF), New York, USA, pp. 374–382, 2023, doi: 10.1145/3604237.3626869.

[11] P. Jiang, C. Xue, Y. Jiang, W. Wei, G. Long, and C. Zhang, “A Survey on Retrieval and Structuring Augmented Generation with Large Language Models,” in Proc. 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), Toronto, Canada, 2025, doi: 10.1145/3715428.

11. APPENDICES

11.1 Survey Instrument

The following survey instrument will be utilized to gather qualitative feedback from domain experts during the system evaluation phase to validate the practical utility of the generated relational metadata.

Section A – Operational Context

1. What is your current role within the AML/Fraud Detection pipeline?
2. How frequently do current rule based systems generate false positives in your daily alert queue?

Section B – Relational Analysis Challenges

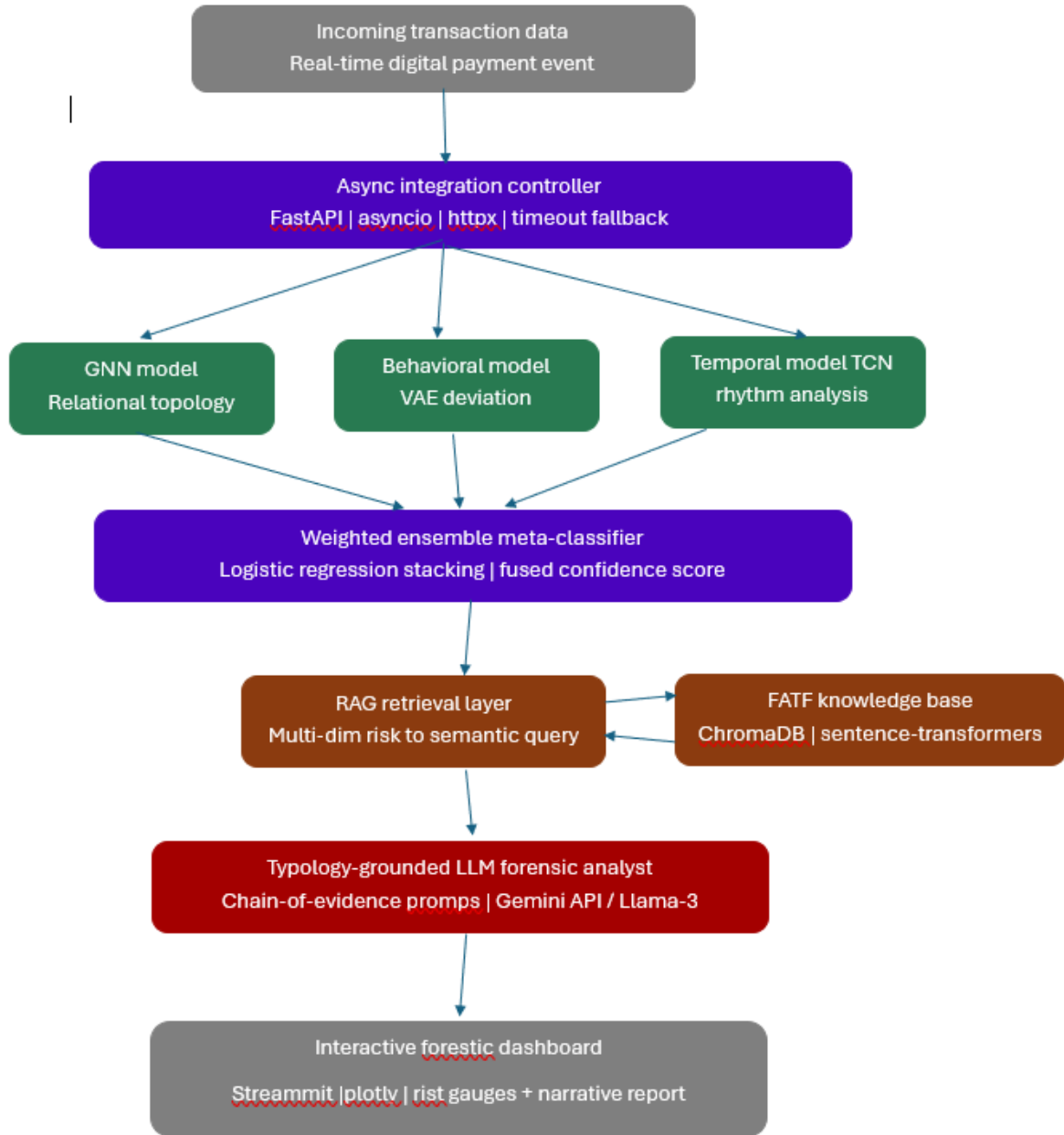
3. When investigating a suspected mule network, how many manual hours does it typically take to trace the funds across multiple associated accounts?
4. How often do isolated threshold alerts miss highly coordinated, slow moving algorithmic fraud?

Section C – System Evaluation (Post Prototype Demo)

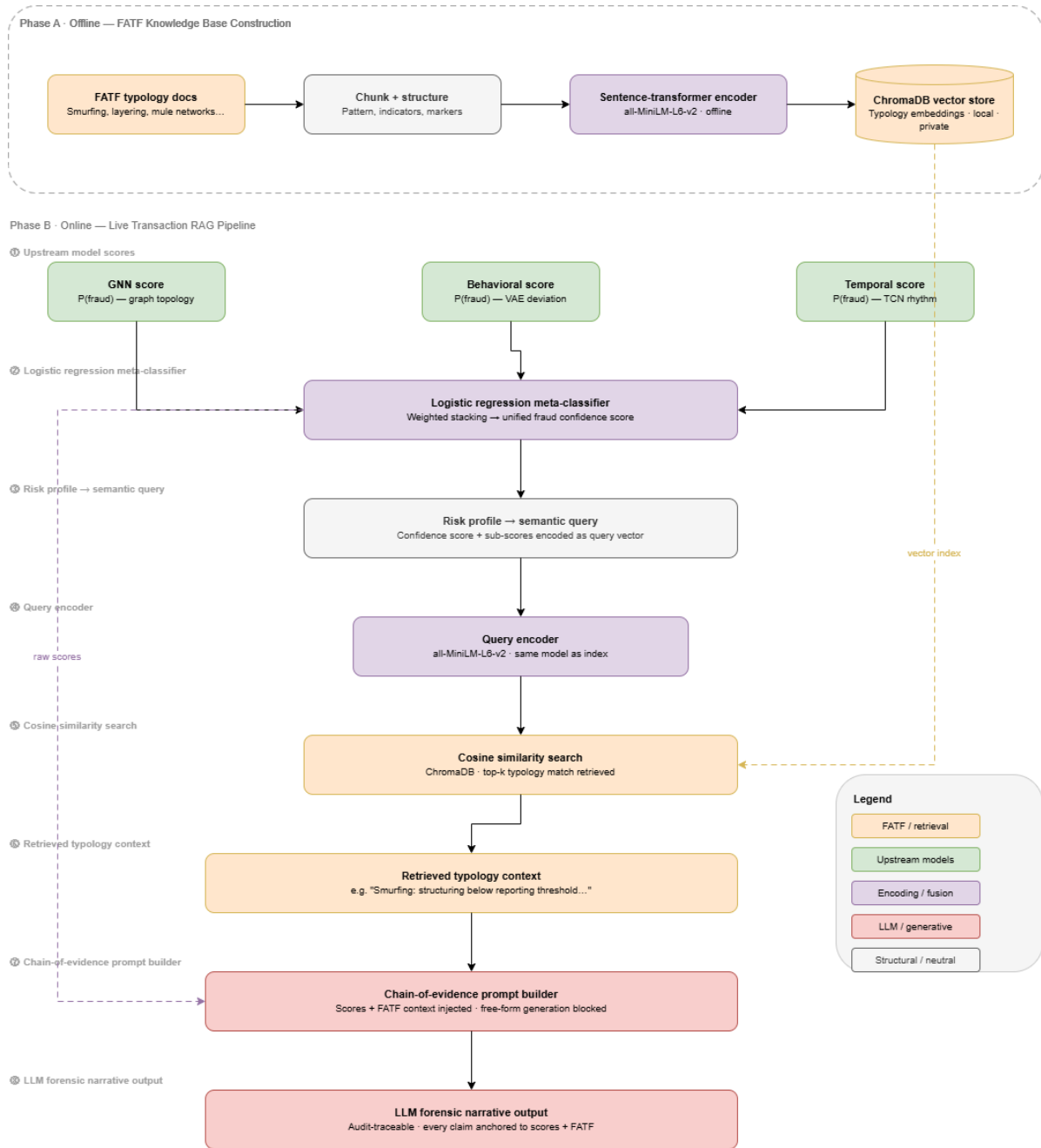
5. Does the inclusion of specific transactional edge attributes (amount, frequency) in the model's reasoning increase your confidence in the alert compared to standard network connection scores?
1. Is the “Suspicious Subgraph” visual output provided by the system effectively communicating the hub and spoke nature of the identified mule ring?
2. Would the relational metadata generated by this system, when passed to a Generative AI for report writing, sufficiently justify an immediate account freeze?

11.2 Detailed diagrams

The following diagrams present the detailed technical architecture and data flow of the proposed **Fusion Engine and Generative Explainability** system.



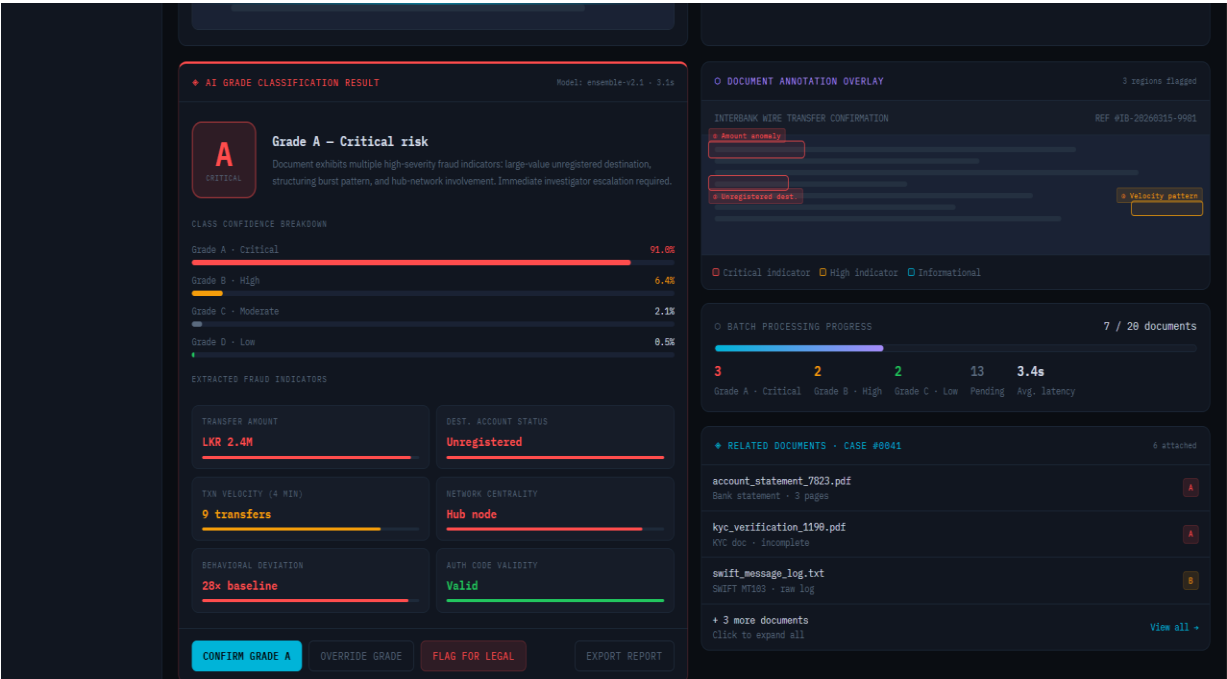
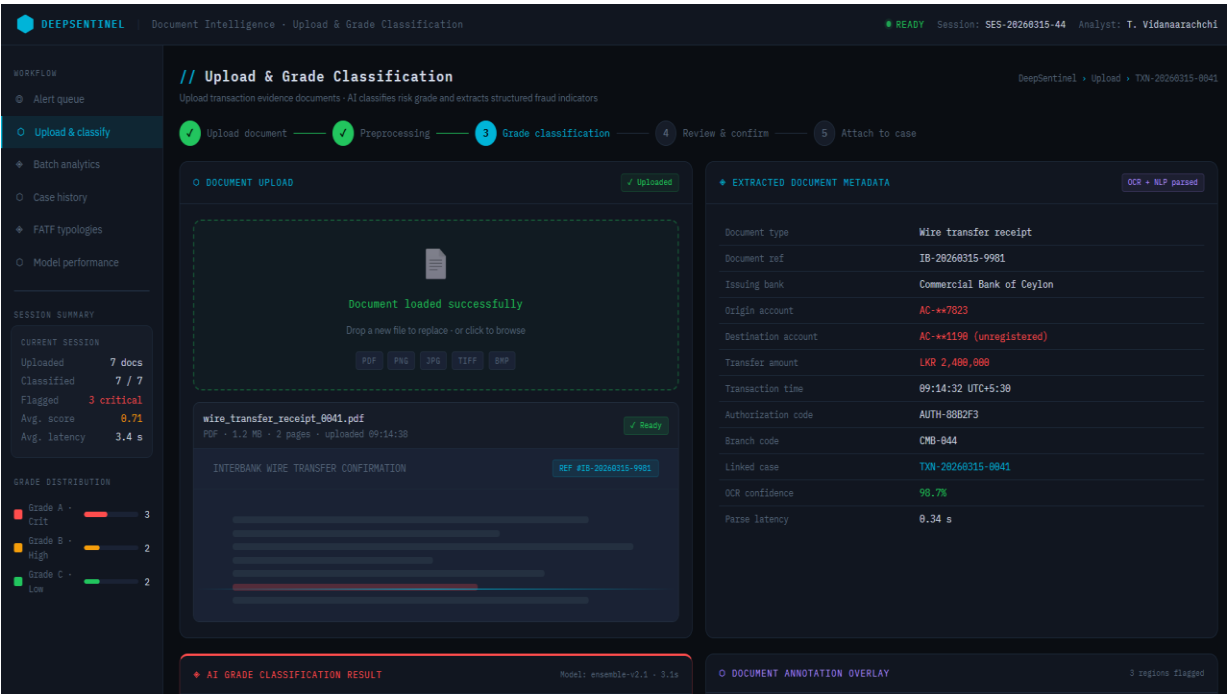
Appendix Figure A1. Detailed System Architecture Diagram



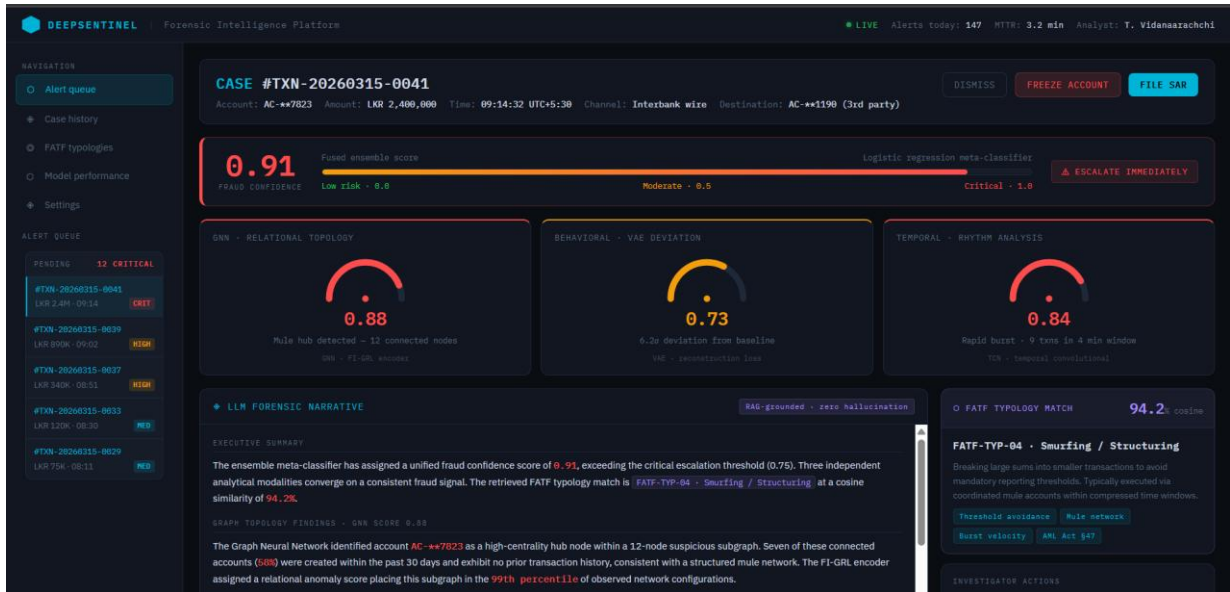
Appendix Figure A2. Detailed Data Flow Diagram

11.3 Preliminary UI mockups

The following UI mockups illustrate the key screens of the proposed prototype interface for the **Fusion Engine forensic dashboard**.



Appendix Figure A3. UI Mockup — Forensic Dashboard Risk Gauge View



Appendix Figure A4. UI Mockup — Batch Analytics Dashboard

11.4 Risk analysis

Risk	Impact	Likelihood	Mitigation Strategy
LLM Generative Hallucination	High	High	Restrict the language model using a strict Chain of Evidence prompt template. The LLM is forced to generate reports citing only the retrieved FATF definitions and the numerical risk arrays, completely disabling free form generation.
Upstream Analytical Model Timeout	High	Medium	Implement asynchronous FastAPI routing with strict timeout thresholds. If a sub model fails to respond, the Integration Controller will proceed with the available scores and explicitly document the missing modality in the final report.

High API Latency Bottlenecking the Dashboard	High	Medium	Utilize native asynchronous Python libraries
Data Privacy Leakage via Public LLM APIs	Medium	low	Architect the system to support fully local, offline execution. Deploy the ChromaDB instance locally and support localized language models (Llama 3 via Ollama) for enterprise testing, ensuring no simulated transaction data leaves the internal network.
Meta Classifier Overfitting to PaySim Synthetics	Medium	Low	Train the Logistic Regression ensemble layer utilizing cross validation on the PaySim validation splits. Maintain a modular architecture so the ensemble weights can be rapidly recalibrated if data drift occurs.
Complete Upstream Dependency Failure (All 3 models unavailable)	High	Low	Maintain the internal “Mock Score Generator” developed during Phase 1. This allows the Fusion Engine and the generative RAG layer to be fully demonstrated and evaluated using simulated inputs even if the upstream research components fail to deliver.

11.5 Ethical documents

Appendix Note 1: Confirmation that the system is trained and evaluated exclusively using the PaySim synthetic dataset, which contains no real Personally Identifiable Information (PII), eliminating all GDPR and financial privacy compliance risks.

Appendix Note 2: Declaration that the FATF fraud typology knowledge base is constructed strictly from publicly available regulatory guidance documents published by the Financial Action Task Force, with no proprietary or confidential data sources.

Appendix Note 3: Data localization guarantee confirming that all ChromaDB vector store processing and LLM inference occur entirely within the local institutional environment during enterprise deployment, ensuring no simulated transaction metadata is transmitted to external public APIs.

Appendix Note 4: Open source compliance declaration confirming that all frameworks used, including ChromaDB, sentence transformers, FastAPI, Streamlit, and Ollama, are utilized under their respective open source academic and commercial licenses.